SMART ATTENDANCE SYSTEM USING FINGERPRINT + KEYPAD BACKUP

ES333 – Microprocessors & Embedded Systems

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Problem Statement

Why this project?

- Manual attendance is slow and inefficient
- Prone to proxy attendance using **biometric attendance**
- Biometric alone can fail (wet/damaged finger)
- Need secure + offline storage + reliable **automated hardware-based solution**
- **Secure data storage** using microcontroller-based control
- Attendance must be exportable to PC

Proposed Solution

What are we building?

- Fingerprint authentication (primary)
- Keypad roll-number fallback
- Admin enrollment mode (PIN protected)
- SD card storage of attendance logs
- PC-readable data (CSV format)
- Fully standalone embedded system

Embedded Design Overview

STM32:

- Fingerprint processing
- Attendance validation
- SD card write operation

Vega:

- LCD display
- Keypad interface
- User feedback

Storage:

- SD card stores attendance logs
- File format: `.csv`
- Can be opened directly in Excel / PC

Hardware Components

Component	Purpose
STM32	Core control + SD handling
Vega	UI handling
Fingerprint Sensor	Biometric input
4x4 Keypad	Roll entry + admin PIN
LCD	Display messages
Buzzer	Feedback
SD Card Module (SPI)	Attendance storage

Working Flow in short

Normal Operation

- Finger placed
- Match found → mark attendance
- Save record to SD card

Format:
`RollNo, Date, Time, Mode`

If Fingerprint Fails

- Prompt for roll number
- Validate entry
- Save as: `MANUAL`

Admin Mode

- Enter enrollment mode
- Enter PIN via keypad
- Capture fingerprint
- Assign roll number
- Store fingerprint template
- Update database

Optional:

- Delete user

Interfaces & Protocols

Interfaces & Protocols

- Fingerprint → UART
- SD Card → SPI
- LCD → I2C
- Keypad → GPIO scanning
- STM32 ↔ Vega → UART

SD Storage Details

- FAT32 formatted card
- File created daily
- Append mode writing
- Safe file closing after each write

THANK YOU